

[HINT SOLUTION]

Q.1 $\frac{1+i}{\sqrt{2}}$

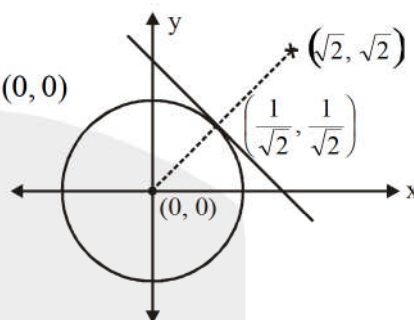
Sol. $\therefore \left| \frac{z - \sqrt{2}(1+i)}{z} \right| = 1 \Rightarrow |z - \sqrt{2}(1+i)| = |z - 0|$

$\therefore z$ is on perpendicular bisector of line joining $(\sqrt{2}, \sqrt{2})$ and $(0, 0)$

$\therefore (0, 0)$ is centre of circle $|z| = 1$

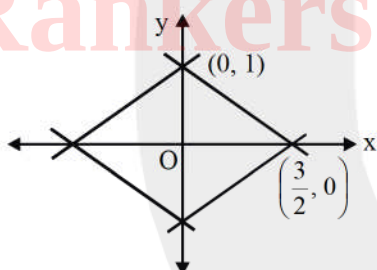
$\therefore$ point z will be $\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right)$

$\therefore z = \frac{1+i}{\sqrt{2}}$



Q.2 3 sq. units

Sol. $2|z + \bar{z}| + 3|z - \bar{z}| = 6$
 $4|x| + 6|y| = 6$
 $2|x| + 3|y| = 3$



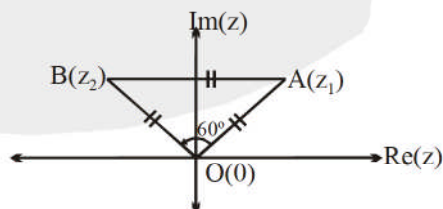
Required area $= 4 \times \frac{1}{2} \times \frac{3}{2} \times 1 = 3$.

Q.3 equilateral

Sol. Given, $\frac{z_2 - 0}{z_1 - 0} = e^{i\pi/3} \Rightarrow \left| \frac{z_2 - 0}{z_1 - 0} \right| = \left| e^{i\pi/3} \right| = 1$

$\Rightarrow |z_2 - 0| = |z_1 - 0|$

Also, $\arg\left(\frac{z_2}{z_1}\right) = \frac{\pi}{3} \Rightarrow \angle z_1 O z_2 = \frac{\pi}{3} \Rightarrow$ Triangle is equilateral. **Ans.**



Aliter :

$\frac{z_2}{z_1} = \frac{1+i\sqrt{3}}{2}$

Subtract 1 on b.t.s., we get

$$\Rightarrow \frac{z_2}{z_1} - 1 = \frac{-1 + i\sqrt{3}}{2} \Rightarrow |z_2 - z_1| = |z_1| = |z_2|$$

$\Rightarrow$ Triangle is equilateral.

Q.4 $\operatorname{cosec} 3^\circ$ (D) $\frac{1}{10} \operatorname{cosec} 3^\circ$

Sol. As, $\frac{1}{z^{2n+1}} = (\cos(2n+1)\theta - i \sin(2n+1)\theta)$

$$\therefore \frac{4i}{z^{2n+1}} = 4i (\cos(2n+1)\theta - i \sin(2n+1)\theta) \Rightarrow \operatorname{Re}\left(\frac{4i}{z^{2n+1}}\right) = 4 \sin(2n+1)\theta$$

$$\begin{aligned} \therefore \sum_{n=0}^9 \operatorname{Re}\left(\frac{4i}{z^{2n+1}}\right) &= 4 \sum_{n=0}^9 \sin(2n+1)\theta = 4 (\sin \theta + \sin 3\theta + \sin 5\theta + \dots + \sin 19\theta) \\ &= 4 \left(\frac{\sin(10\theta) \cdot \sin(10\theta)}{\sin \theta} \right) = 4 \times \frac{\sin^2(30^\circ)}{\sin 3^\circ} = \operatorname{cosec} 3^\circ. \end{aligned}$$

Q.5 $\frac{1}{2\sqrt{3}}$

Sol. Given $|z-1| = 2 \operatorname{Im}(z)$

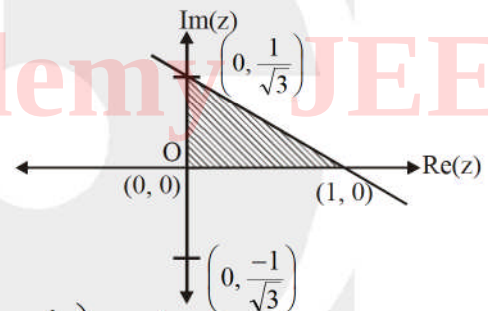
$$\therefore (x-1)^2 + y^2 = 4y^2 \Rightarrow 3y^2 = (x-1)^2$$

$$\Rightarrow \pm \sqrt{3}y = (x-1)$$

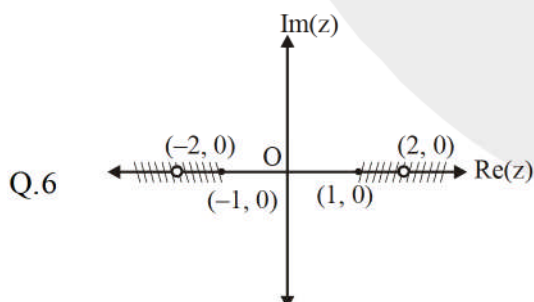
$$\therefore L_1: x - \sqrt{3}y - 1 = 0 \quad (\text{Rejected}) \text{ think?}$$

$$\text{and } L_2: x + \sqrt{3}y - 1 = 0.$$

$$\text{Also, } L_3: x = 0 \text{ and } L_4: y = 0$$



$$\text{So, area of triangle formed by } L_2, L_3 \text{ and } L_4 \text{ is } = \left(\frac{1}{2} \times 1 \times \frac{1}{\sqrt{3}} \right) = \frac{1}{2\sqrt{3}}.$$



Sol. Consider $\frac{-4iz}{4-z^2} = \frac{(-4i)(2e^{i\theta})}{4-4e^{i2\theta}}$ (As $|z| = 2 \Rightarrow z = 2e^{i\theta}$)

$$= \frac{-2i(\cos \theta + i \sin \theta)}{(1 - \cos 2\theta) + (-i \sin \theta)} = \frac{-2i(\cos \theta + i \sin \theta)}{-2i^2 \sin^2 \theta - 2i \sin \theta \cdot \cos \theta}$$

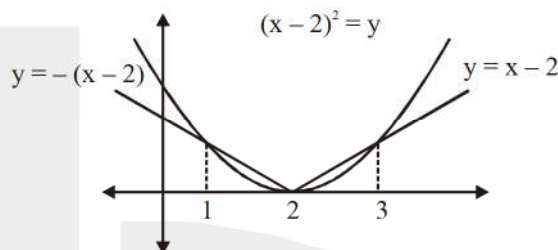
$$= \frac{\cos \theta + i \sin \theta}{\sin \theta (\cos \theta + i \sin \theta)} = \operatorname{cosec} \theta \in (-\infty, -1] \cup [1, \infty) \Rightarrow \text{Option (B) is correct.}$$

Q.7 $\frac{1}{3}$

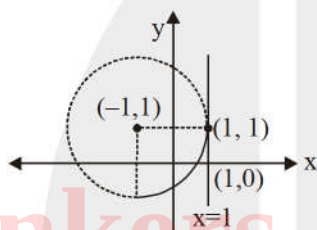
Sol. $A = 2 \int_2^3 (x-2 - (x-2)^2) dx$

$$= 2 \left[\frac{(x-2)^2}{2} - \frac{(x-2)^3}{3} \right]_2^3$$

$$= 2 \left[\frac{1}{2} - \frac{1}{3} \right] = \frac{1}{3}.$$



Q.8 1
Sol. $z = 1 + i$, only satisfies both.



Q.9 25

Hint: $||z_1| - |z_2|| \leq |z_1 + z_2|$ (Using left side of triangle inequality.)
Here, $1 \leq |z| \leq 25$.

Q.10 $\sqrt{3}$

Sol. Let $z = x + iy$
 $-4y^2 = 4(x^2 + y^2) - 12$

$$\Rightarrow x^2 + 2y^2 = 3 \Rightarrow \frac{x^2}{3} + \frac{y^2}{3/2} = 1$$

$$x = \sqrt{3} \cos \theta, y = \sqrt{\frac{3}{2}} \sin \theta$$

$$x^2 + y^2 = 3 \cos^2 \theta + \frac{3}{2} \sin^2 \theta = \frac{3}{2} (2 \cos^2 \theta + \sin^2 \theta) = \frac{3}{2} (1 + \cos^2 \theta)$$

$$|z|^2 \Big|_{\max} = \frac{3}{2} \times 2 = 3$$

$$\Rightarrow |z|_{\max} = \sqrt{3}$$



Q.11 4k

Sol. [D]

n^{th} root of unity

$$\cos \frac{2m\pi}{n} + i \sin \frac{2m\pi}{n}$$

Let for $m = k$, root of unity makes an angle of 90°

$$\text{with } \frac{2k\pi}{n} = \frac{\pi}{2} \Rightarrow n = 4k$$

for $m = 0$

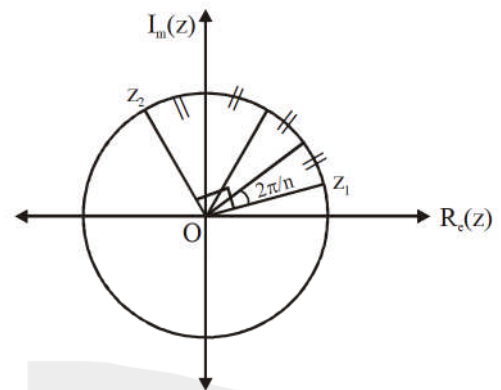
$$z_1 = 1$$

for $m = k$

$$z_2 \text{ such that } z_2 O z_1 = 90^\circ$$

Aliter :

$$\therefore \left(\frac{2\pi}{n} \right) k = \frac{\pi}{2} \Rightarrow n = 4k$$



Q.12 $5\sqrt{2}$

Q.13 $-2 + \sqrt{2}$

Q.14 4 sq. units

Sol. Using triangle inequality

$$|(z^2 - 2iz) - (z^2 - 9z + 7iz)| \leq 18$$

$$|9z - 9iz| \leq 18 \Rightarrow |z| \leq \sqrt{2}$$

$$C_1 : x^2 + y^2 = 2$$

$$C_2 : \frac{x^2}{2} - \frac{y^2}{16} = 1$$

$$(i) |z_0 + 4 + 4i| \leq |z_0| + |4 + 4i|$$

$$\leq \sqrt{2} + 4\sqrt{2}$$

$$\leq 5\sqrt{2}$$

$$(ii) z_1 = \sqrt{2}$$

$$z_2 = -1 + i$$

$$z_3 = \bar{z}_2 = -1 - i$$

$$\therefore \operatorname{Re}(z_1 + z_2 + z_3) = -2 + \sqrt{2}$$

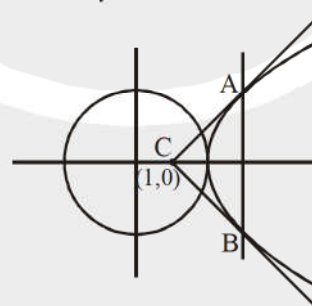
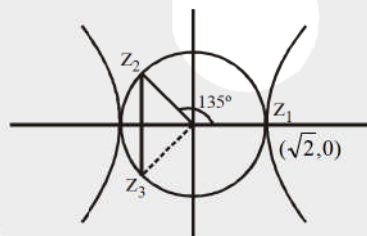
(iii) Line AB

$$y = 4x - 4$$

which is tangent to the curve C_2 .

$$A \equiv (2, 4), B \equiv (2, -4) \text{ and } C \equiv (1, 0)$$

$$\therefore \operatorname{Ar}(\triangle ABC) = \frac{1}{2} \times 1 \times 8 = 4 \text{ sq. units.}$$



Q.15 (B) area of the triangle formed by the straight line $x - 3y + 1 = 0$ and tangent at one of the vertices and

the x-axis is $\frac{3}{2}$ sq. units.

(C) distance between directrices of the curve 'C' is $\frac{18}{\sqrt{10}}$.

(D) equation of the circle passing through the points (2, 1) and both foci of the curve 'C' is $x^2 + y^2 + 2x - 9 = 0$.

Sol. ^{40570/cn/MORE} Locus of z is

$$\frac{(x+1)^2}{9} - \frac{y^2}{1} = 1$$

Now verify all options.

Q.16 (A) $a + b = 18$ (C) $a - b = 4\sqrt{2}$ (D) $ab = 73$

Sol. Centre of circle is $(-4, 5)$.

$$\text{Also, radius} = \sqrt{(-4)^2 + (5)^2 + 40} = 9$$

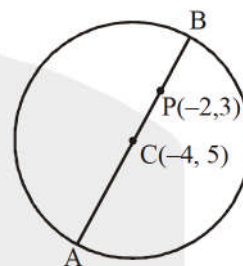
$$\therefore \text{Distance of centre } (-4, 5) \text{ from } (-2, 3) = 2\sqrt{2}$$

$$\text{So, } a = \max. |z - (-2 + 3i)| = 9 + 2\sqrt{2}$$

$$\text{and } b = \min. |z - (-2 + 3i)| = 9 - 2\sqrt{2}$$

$$\text{Hence, } a + b = 18$$

$$\text{Also, } (a - b) = 4\sqrt{2} \text{ and } ab = \frac{(a+b)^2 - (a-b)^2}{4} = \frac{324 - 32}{4} = \frac{292}{4} = 73.$$



Q.17 11

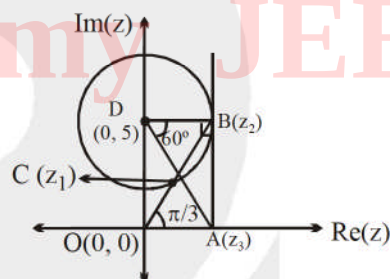
Sol. As, O, A, B and D are concyclic,

$$\text{so } \cos 60^\circ = \frac{BD}{AD} = \frac{3}{AD}$$

$$\therefore AD = 6 \text{ and } OD = 5$$

$$\text{Hence, } |z_3| = OA = \sqrt{36 - 25} = \sqrt{11}$$

$$\text{Hence } |z_3|^2 = 11.$$

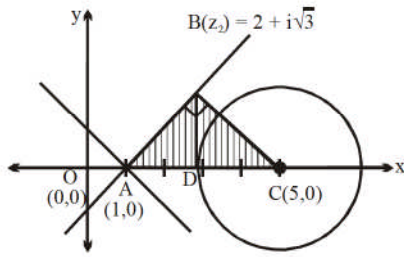


Q.18 2

$$\text{Sol. } S = \frac{\sum_{k=1}^{12} |\bar{\alpha}_k + \alpha_{12-k}|}{\sum_{k=1}^6 |\alpha_{2k-1} + \alpha_{2k}|} = \frac{\sum_{k=1}^{12} |\alpha_{13-k} + \alpha_{12-k}|}{\sum_{k=1}^6 |\alpha_{2k-1}(1 + \alpha_1)|} = \frac{\sum_{k=1}^{12} |\alpha_{12-k}(\alpha_1 + 1)|}{\sum_{k=1}^6 |\alpha_{2k-1}| |1 + \alpha_1|} = \frac{\sum_{k=1}^{12} |\alpha_1 + 1|}{\sum_{k=1}^6 |\alpha_1 + 1|} = 2$$

Q.19 12

Sol.



Clearly $A(z_1)$ is the point of intersection of $\arg(z-2+i) = \frac{3\pi}{4}$ and $\arg(z+\sqrt{3}i) = \frac{\pi}{3}$
 $\Rightarrow z_1 = 1$

Also, $B(z_2)$ is the point on $\arg(z+\sqrt{3}i) = \frac{\pi}{3}$ such that $|z_2-5|$ is minimum, so $z_2 = 2+i\sqrt{3}$.

also, $C(z_3)$ be the centre of the circle $|z-5|=3$, so $z_3 = 5$.

Hence, area of $\triangle ABC = \frac{1}{2}(AC) \times (BD) = \frac{1}{2}(4) \times (\sqrt{3}) = 2\sqrt{3}$ (square unit.)

$$\therefore \Delta = 2\sqrt{3} \Rightarrow \Delta^2 = 12$$

Q.20 136

Sol. It is to be noted that two of the numbers need to be conjugates and one number must be real, as the coefficients of the cubic are all real.

Three roots are, $\omega + 3i$, $\omega + 9i$ and $2\omega - 4$

let $\omega + 3i$ is real, hence $\omega = \alpha - 3i$ where $\alpha \in \mathbb{R}$

then $(\alpha - 3i + 9i)$ and $(2\alpha - 6i - 4)$

i.e. $\alpha + 6i$ and $(2\alpha - 4) - 6i$ must be complex conjugate $\Rightarrow \alpha = 2\alpha - 4$

$\therefore \alpha = 4$

Hence the roots are

$4, 4 + 6i, 4 - 6i$ (other options not possible)

the equation is

$$(z-4)(z^2-8z+5z) \Rightarrow z^3-12z^2+84z-208$$

Hence $|a+b+c| = |-12+84-208| = 136$.